NPTEL : Electronic Devices
Lab No. 1 & 2

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1 Lab 1: Diode I-V Characterization and Band Gap of Semiconductors

1.1 Aim

To study the forward bias I-V characteristics of a silicon diode (1N914) and LEDs of different colours (Red, Green, Blue and White), determine their cut-in voltages, estimate their band gaps, calculate ideality factor and saturation current, and study the correlation between cut-in voltage and semiconductor band gap.

1.2 Observations

White LED

Current (mA)	Voltage (V)
0.072	2.53
0.747	2.62
1.00	2.65
2.98	2.73
5.13	2.80
8.78	2.90

Table 1: White LED I-V Measurements

Green LED

Current (mA)	Voltage (V)
0.32	2.27
0.512	2.31
1.00	2.38
2.04	2.48
2.88	2.55
8.94	2.87

Table 2: Green LED I-V Measurements

Red LED

Current (mA)	Voltage (V)
0.244	1.74
1.00	1.78
2.02	1.83
4.14	1.87
8.38	1.94
13.53	1.99

Table 3: Red LED I-V Measurements

Blue LED

Current (mA)	Voltage (V)
0.005	2.43
0.183	2.54
1.00	2.62
2.78	2.69
4.92	2.78
8.78	2.89

Table 4: Blue LED I-V Measurements

1N914 Silicon Diode

Current (mA)	Voltage (V)
0.00	0.10
1.00	0.59
1.87	0.62
4.60	0.67
10.29	0.72
19.93	0.76

Table 5: 1N914 I-V Measurements

1.3 Plots

I-V Characteristics of all Diodes

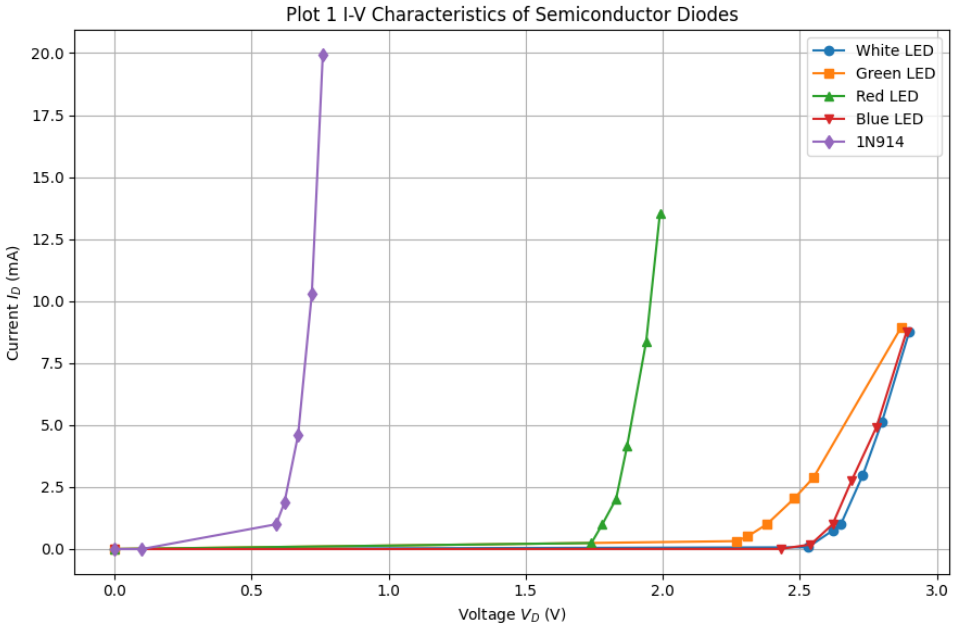


Figure 1: I-V Characteristics of 1N914, Red, Green, Blue and White LEDs

Semi-logarithmic Characteristics

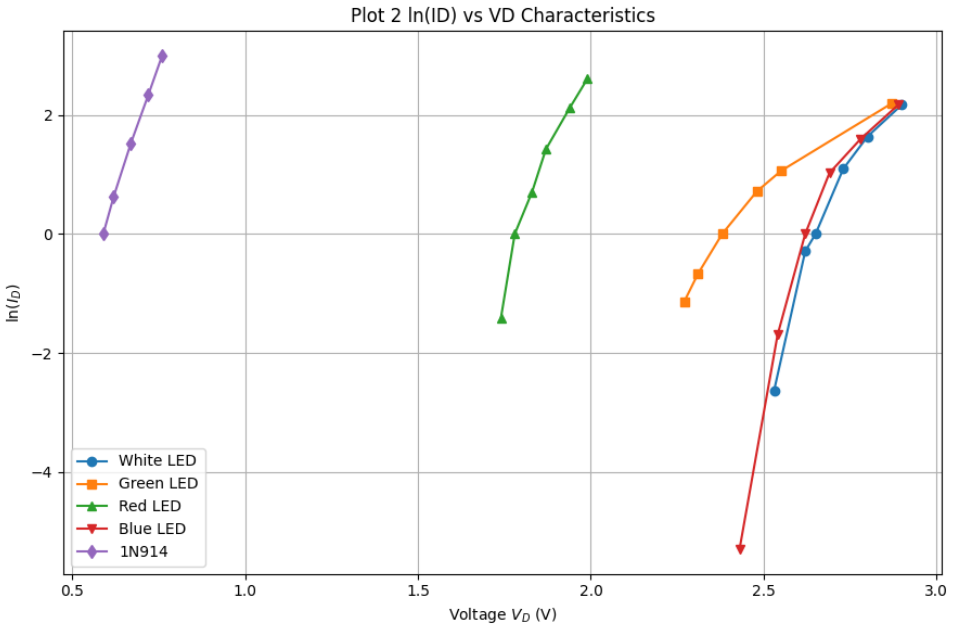


Figure 2: $\ln(I_D)$ versus V_D Characteristics

Cut-in Voltage versus Band Gap

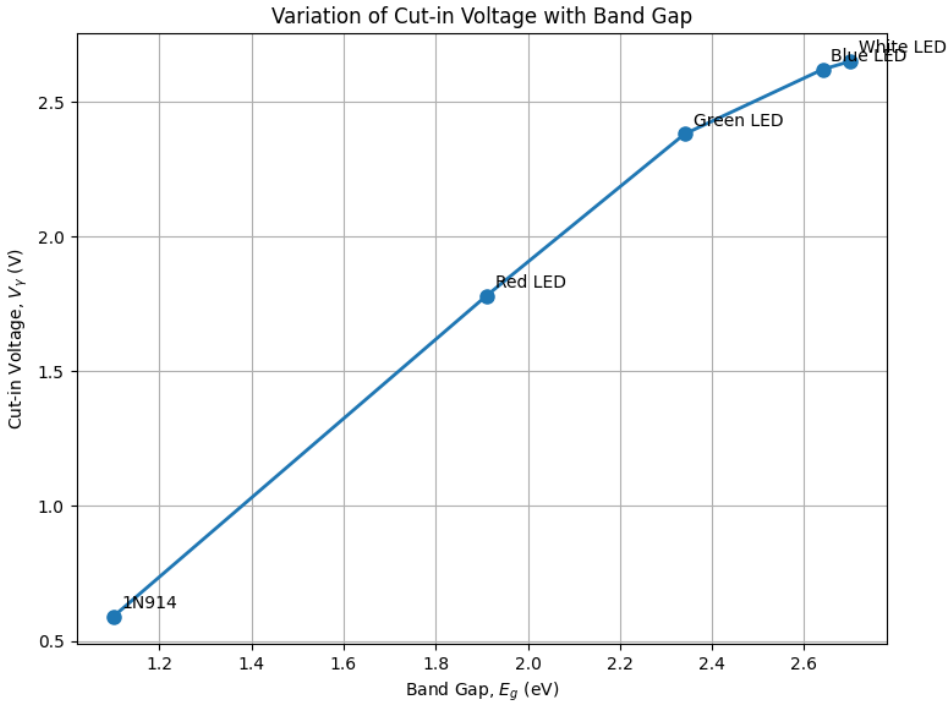


Figure 3: Variation of Cut-in Voltage with Band Gap

1.4 Results

Diode	Eg (eV)	VTH (V)	Ideality Factor	Is (mA)
1N914	1.10	0.59	2.21	3.64×10^{-5}
Red LED	1.91	1.78	2.55	1.65×10^{-12}
Green LED	2.34	2.38	7.16	2.39×10^{-6}
Blue LED	2.64	2.62	2.46	6.56×10^{-19}
White LED	2.70	2.65	3.12	4.61×10^{-15}

Table 6: Summary of Experimental Results

1.5 Answers to Questions

Correlation between V_{TH} and E_g

The cut-in voltage increases with increasing band gap. Materials having larger band gaps require higher forward voltages for conduction.

Validity of Diode Equation

The diode equation is valid only in the exponential region. At low currents leakage current dominates, while at high currents series resistance causes deviation from ideal behaviour.

Effect of Choosing Different Current Levels

At low currents measurement errors and leakage currents affect the readings. At high currents series resistance increases the forward voltage. Therefore, values around 1 mA provide the most reliable estimate.

Intrinsic Carrier Density
The intrinsic carrier density decreases exponentially with increasing band gap.

Applications
1N914

- High-speed switching circuits
- Clipping and clamping circuits
- Logic circuits

Red LED

- Status indicators
- Seven-segment displays
- Signboards

Green LED

- Traffic signals
- Instrument indicators
- Display systems

Blue LED

- Full-colour displays
- Blu-ray storage systems
- Decorative lighting

White LED

- General illumination
- Flashlights
- LCD backlighting
- Street lighting

1.6 Conclusion

The forward bias I-V characteristics of the silicon diode and LEDs were studied successfully. The cut-in voltage increased with increasing band gap. The semi-logarithmic characteristics verified the exponential relationship between current and voltage predicted by the diode equation.

2 Lab 2: I-V Characteristics of Fabricated Metal-Semiconductor Diode (N-15)

2.1 Aim

To study the forward and reverse bias I-V characteristics of the fabricated metal-semiconductor diode sample (N-15).

2.2 Observations

Forward Bias Characteristics

Voltage (V)	Current (mA)
0.0	0.000
0.1	0.000
0.2	0.013
0.3	0.027
0.4	0.037
0.5	0.044
0.6	0.052
0.7	0.058
0.8	0.064
0.9	0.072
1.0	0.078
1.1	0.082
2.0	0.132
2.5	0.158
3.0	0.183
3.5	0.214

Table 7: Forward Bias I-V Characteristics of N-15 Sample

Reverse Bias Characteristics

Voltage (V)	Current (mA)
0.0	0.000
0.1	0.001
0.2	0.001
0.3	0.001

Table 8: Reverse Bias I-V Characteristics of N-15 Sample

2.3 Results

- Forward current increased gradually with increasing applied voltage.
- Current conduction became noticeable beyond approximately 0.2 V.

- Reverse current remained very small (approximately 0.001 mA).
- The measured I-V characteristic exhibited rectifying behaviour.

2.4 Conclusion

The forward and reverse bias I-V characteristics of the N-15 fabricated sample were measured successfully. A maximum forward current of 0.214 mA was observed at 3.5 V, while the reverse current remained approximately 0.001 mA within the measured range.